

Two-Dimensional Ferroelectric Altermagnets: From Model to Material Realization

Ziye Zhu,[⊥] Xunkai Duan,[⊥] Jiayong Zhang,[⊥] Bowen Hao, Igor Žutić, and Tong Zhou*



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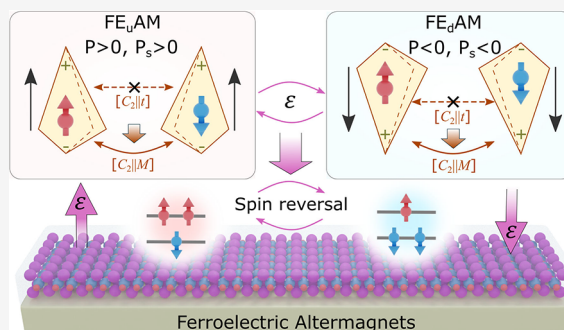
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Supporting Information

ABSTRACT: Multiferroic altermagnets offer new opportunities for magnetoelectric coupling and electrically tunable spintronics. However, due to intrinsic symmetry conflicts between altermagnetism and ferroelectricity, achieving their coexistence, known as ferroelectric altermagnets (FEAM), remains an outstanding challenge, especially in two-dimensional (2D) systems. Here, we propose a universal, symmetry-based design principle for 2D FEAM, supported by tight-binding models and first-principles calculations. We show that lattice distortions can break the spin equivalence and introduce the necessary rotation-related symmetry, enabling altermagnetism with electrically reversible spin splitting. Guided by this framework, we identify a family of 2D vanadium oxyhalides and sulfide halides as promising FEAM candidates. In these compounds, pseudo-Jahn–Teller distortions and Peierls-like dimerization cooperatively establish the required symmetry conditions. We further propose the magneto-optical Kerr effect as an experimental probe to confirm FEAM and its electric spin reversal. Our findings provide a practical framework for 2D FEAM and advancing electrically controlled spintronic devices.

KEYWORDS: altermagnets, multiferroics, ferroelectrics, spintronics, 2D materials



Electric control of magnetism remains a long-standing goal in multiferroics and spintronics, with the potential to revolutionize high-density data storage and energy-efficient spintronic devices.^{1–4} Multiferroic materials, which exhibit coexisting magnetic and (anti)ferroelectric orders, are particularly enticing for this purpose, because they couple electric and magnetic properties within a single phase. Recent discoveries of altermagnets (AM),^{5–25} a new class of collinear magnets that display spin splitting without net magnetization, further expand this research frontier. Integrating (anti)-ferroelectricity with altermagnetism offers a pathway to fast, reversible control of magnetism and spin polarization.^{26–32}

A key challenge in realizing such multiferroic altermagnets lies in establishing the appropriate symmetry, particularly rotation-related symmetry, R , that connects opposite-spin sublattices.^{6,26} While conventional ferroelectricity often preserves translation symmetry, t , leading to traditional antiferromagnets (AFM), antiferroelectricity naturally introduces the R essential for AM.²⁶ This has recently led to the concept of antiferroelectric altermagnets, in which an electric field toggles spin polarization on and off by switching between antiferroelectric and ferroelectric phases.²⁶ While antiferroelectric altermagnets have been realized in both three-dimensional (3D) perovskites and two-dimensional (2D) van der Waals materials, ferroelectric altermagnets (FEAM) are exceedingly rare, with only two switchable 3D FEAM identified across the entire MAGNDATA database.²⁷

Despite their promise, FEAM research has so far been limited to 3D bulk crystals.^{27,28} Extending FEAM to 2D systems is both timely and essential for the development of nanoscale devices, in which atomically thin materials offer key advantages such as improved scalability, tunable physical properties, and facile integration with flexible electronics.³³ However, 2D FEAM remain largely unexplored, and a general theoretical framework that explains how ferroelectricity can induce altermagnetism in 2D systems is still lacking. This absence not only limits the fundamental understanding of FEAM but also hinders systematic materials discovery. Given the rising demand for ultrathin materials with electrically switchable magnetism, identifying and designing a 2D FEAM have become urgent goals.

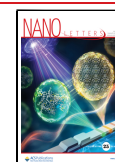
In this work, we overcome these challenges by introducing a universal design principle for 2D FEAM and confirming its feasibility through both model analysis and first-principles calculations. We first construct a tight-binding (TB) model showing how lattice distortions break translation symmetry to enable spin-inequivalent hopping, triggering altermagnetism.

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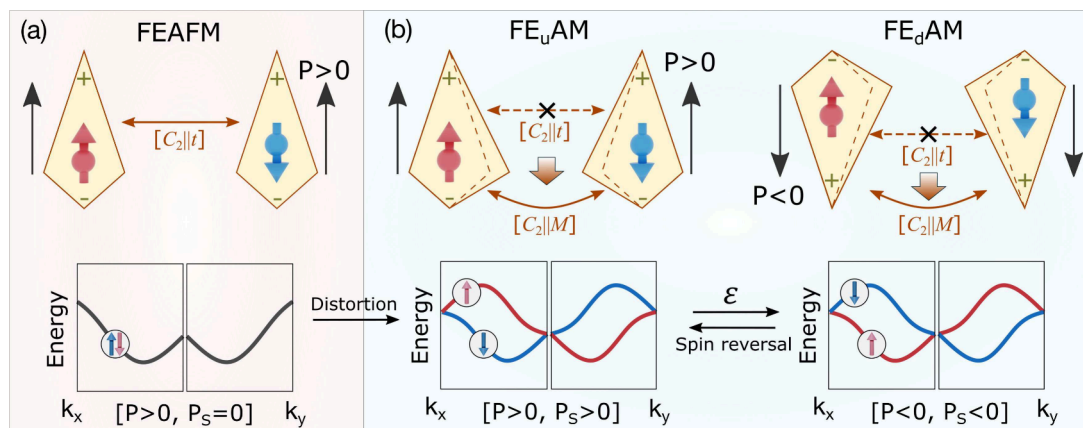


Figure 1. Design principle for FEAM. Magnetic atoms with opposite spins (red and blue arrows) form AFM sublattices, while parallel local electric polarizations (black arrows), arising from asymmetric charge distributions, produce a FE state with non-zero net electric polarization, P . (a) For FEAFM, magnetic sublattices are directly connected by translation symmetry, t , resulting in conventional AFM with a vanishing spin splitting ($P_s = 0$). (b) For FEAM, lattice distortions break t but preserve mirror symmetry, M , between magnetic sublattices, creating AM with $P_s > 0$. Depending on the P direction, two distinct states emerge: FE_uAM with $[P > 0, P_s > 0]$ and FE_dAM with $[P < 0, P_s < 0]$. An electric field, ϵ , can switch between the FE_uAM and FE_dAM states, enabling electric control of P_s .

Crucially, reversing the FE polarization flips the spin splitting by interchanging sublattice-dependent hoppings. Guided by this model, we identify a family of well-established 2D materials, vanadium oxyhalides and sulfide halides (VOX_2 and VSX_2 , respectively, with $X = \text{Cl}, \text{Br}, \text{or I}$),^{34–37} as promising FEAM candidates. In these compounds, ferroelectricity arises from pseudo-Jahn–Teller distortions, while Peierls-like dimerization induces the R symmetry to realize the FEAM. Our simulations confirm the reversibility of the ferroelectrically-controlled spin splitting, and we propose the magneto-optical Kerr effect (MOKE) as a viable experimental probe. By integrating symmetry analysis, an effective model, and first-principles validation, our work not only establishes a promising pathway for achieving 2D FEAM but also lays the groundwork for their future experimental realization and spintronic applications.

According to spin group theory,^{38–41} realizing AM requires that the magnetic sublattices be related by R rather than by direct t or inversion symmetry.⁶ In typical FE systems, although inversion symmetry is inherently broken, the magnetic sublattices are usually connected by t symmetry. This leads to conventional ferroelectric antiferromagnets (FEAFM), characterized by the spin group $[C_2||t]$, as illustrated in Figure 1a. This incompatibility, where FE states typically preserve t symmetry connecting AFM sublattices, whereas AM demands R symmetry, severely restricts the number of viable FEAM species, as reflected by the fact that only two candidates are identified in the MAGNDATA database.

Overcoming this symmetry conflict to realize FEAM requires the introduction of R without compromising ferroelectricity. We propose achieving this by modifying the local environment of magnetic atoms through targeted lattice distortions such as Jahn–Teller effects, Peierls-like dimerization, and external-stimulus-induced structure deformation.⁴² These distortions break t without disrupting the FE order when they occur perpendicular to P , leading to the emergence of M , a specific form of R , which connects the magnetic sublattices. This transformation changes the spin group from $[C_2||t]$ to $[C_2||M]$, resulting in AM with a finite P_s , as shown in Figure 1b. Remarkably, the sign of P_s is strongly coupled to the

direction of P . As the system transitions between two opposite FE configurations, FE_uAM and FE_dAM , P_s is reversed accordingly. This robust magnetoelectric coupling allows for nonvolatile electric control of spin polarization without altering the Néel vector, offering a promising pathway for low-power spintronic and multiferroic applications.

To further reveal the mechanism of our proposed FEAM and its P -controlled spin reversal, we construct an effective TB model based on a general 2D rectangular lattice with AFM order as in Figure 2. The model incorporates up to fourth-nearest-neighbor hopping, enabling the description of the influence of P and structural distortions. The Hamiltonian takes the form

$$H = \left\{ \sum_{i,j} (f_i^\eta c_i^\dagger c_{i+\eta_j} + g_i^{\kappa_j} c_i^\dagger c_{i+\kappa_j} + h_i^{\delta_j} c_i^\dagger c_{i+\delta_j} + l_i^{\gamma_j} c_i^\dagger c_{i+\gamma_j}) + \text{H.C.} \right\} + M_{A,B} \sum_{i \in A,B} c_i^\dagger \sigma_z c_i \quad (1)$$

where $c_i^\dagger$ and c_i represent the electron creation and annihilation operators at site i , respectively, and σ denotes the Pauli matrix. f_i^η , $g_i^{\kappa_j}$, $h_i^{\delta_j}$, and $l_i^{\gamma_j}$ describe the electron hopping between site i and its first-nearest (NN), second-nearest (2NN), third-nearest (3NN), and fourth-nearest neighbors (4NN), connected by vectors η_j , κ_j , δ_j , and γ_j , respectively. The AFM exchange field is characterized by $M_{A,B}$ on sublattices A and B , where $M_A = -M_B$.

AM arises from the inequivalence of intraspin hoppings between magnetic sublattices.^{6,26} Our TB model reveals that the NN, 2NN, and 3NN hoppings do not contribute to AM, as they fail to generate the required spin inequivalence, even in the presence of lattice distortions, as discussed in Supplemental Note S1. In contrast, 4NN hoppings are sensitive to the lattice distortion and play a critical role in enabling AM. In the undistorted structure (Figure 2a), the spin group remains $[C_2||t]$, ensuring sublattice equivalence ($l_A^{1-4} = l_B^{1-4}$), resulting in spin-degenerate AFM (Figure 2d). Upon introducing lattice distortions that lower the symmetry from $[C_2||t]$ to $[C_2||M]$, the 4NN hoppings become sublattice-inequivalent, i.e., $l_A^{1,3} =$

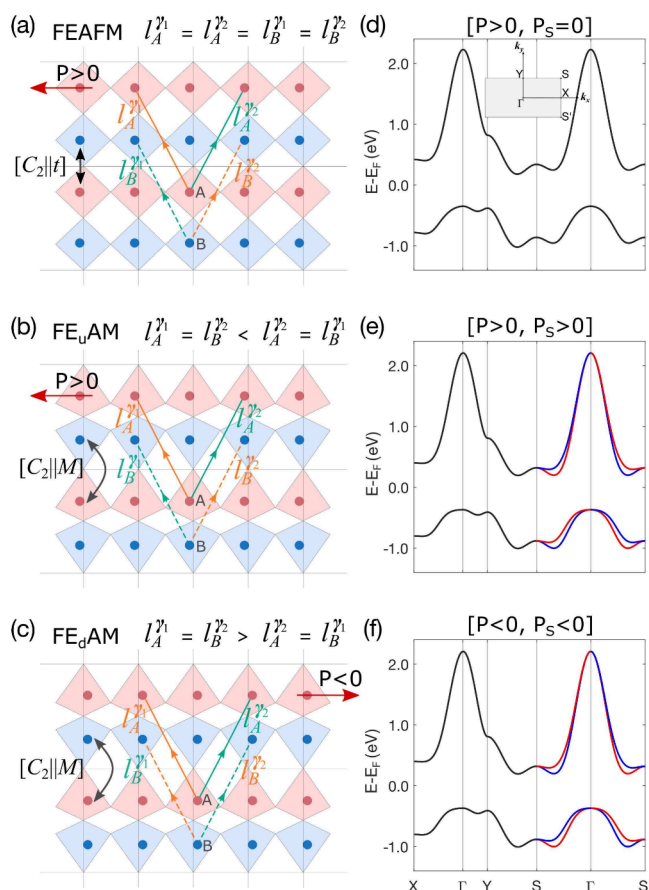


Figure 2. TB model for FEAM. Schematic of a 2D rectangular AFM lattice with FE configurations and lattice distortions. Atomic sites with opposite spins (red and blue dots) form magnetic sublattices A and B. Displacements of atoms from their centrosymmetric positions induce a net $\mathbf{P}$. (a) Undistorted structure with $[C_2||t]$ symmetry gives FEAAM. (b and c) Distorted structure holds $[C_2||M]$ symmetry, resulting in FE_uAM and FE_dAM, respectively. The solid (dashed) arrows indicate 4NN hopping vectors $\gamma_{1,2}$ in the A(B) sublattice with hopping strengths of $t_{A,B}^{1,2}$ ($t_{A,B}^{3,4}$). The other two hopping vectors ($\gamma_{3,4} = -\gamma_{1,2}$) are not shown as the hopping strengths along opposite directions are identical ($t_{A,B}^{3,4} = t_{A,B}^{1,2}$). (d–f) Band structures calculated using the TB model, where the 4NN hopping parameters are (d) $t_{A,B}^{1,2} = t_{A,B}^{3,4} = 0.04$ eV, (e) $t_{A,B}^{1,2} = t_{A,B}^{3,4} = 0.02$ eV and $t_{A,B}^{2,4} = t_{A,B}^{1,3} = 0.06$ eV, and (f) $t_{A,B}^{1,2} = t_{A,B}^{3,4} = 0.06$ eV and $t_{A,B}^{2,4} = t_{A,B}^{1,3} = 0.02$ eV. The inset in panel d illustrates the Brillouin zone and high-symmetry points. For more details about TB parameters, see [Supplemental Note S1](#).

$t_{A,B}^{2,4} < t_{A,B}^{1,3}$ (Figure 2b), giving rise to FEAM, as confirmed by the spin-polarized bands in Figure 2e. Notably, reversing $\mathbf{P}$ interchanges the hopping parameters as $t_{A,B}^{1,3} = t_{A,B}^{2,4} > t_{A,B}^{2,4} = t_{A,B}^{1,3}$ (Figure 2c) and thus reverses the $\mathbf{P}_S$ as in Figure 2f.

These TB model results offer critical insight into the underlying mechanism of our FEAM design. They clearly show how specific lattice distortions can preserve ferroelectricity while breaking translation symmetry, thereby inducing the spin inequivalence necessary for realizing AM. Notably, the model also reveals that reversing $\mathbf{P}$ naturally interchanges the hopping amplitudes between spin sublattices, leading to a $\mathbf{P}_S$ reversal. This makes FEAM a compelling candidate for next-generation spintronic devices. Furthermore, the required lattice distortions can arise from intrinsic mechanisms such as Jahn–Teller effects and Peierls-like dimerization or be externally controlled via temperature, pressure, or chemical doping.⁴² By

harnessing such structural distortions to manipulate magnetic and spin properties, our model not only establishes a foundational framework for FEAM but also provides a versatile strategy for designing and exploring its potential in functional device applications.

We now transition from theoretical design principles to their realization in practical materials. Based on our symmetry analysis and model construction, three key conditions are required to realize FEAM: (1) antiferromagnetic order, (2) ferroelectricity, and (3) R -related symmetry instead of direct t symmetry connecting the magnetic sublattices. These criteria offer a universal framework for the discovery and design of FEAM materials. Guided by this principle, we identify a family of 2D vanadium oxyhalides and sulfide halides (VOX₂ and VSX₂, respectively, where X = Cl, Br, or I)^{34–37} as promising FEAM candidates. Monolayer VOX₂ compounds have been extensively studied as 2D multiferroics that violate the conventional d⁰ rule, exhibiting strong polarizations and magnetoelectric coupling.³⁴ In this work, we focus on monolayer VOI₂ to demonstrate the realization of FEAM, while results for other members of the family are provided in the [Supporting Information](#). VOI₂ exhibits two structural phases, an undistorted phase (Figure 3a) and a distorted phase (Figure 3b),^{34–37} which correspond directly to the model structures in panels a and b, respectively, of Figure 2. Our energy calculations confirm that VOI₂ adopts an antiferromagnetic ground state (Figure 3b), as reported in the previous work,³⁷ with detailed comparisons of various magnetic configurations presented in Figure S3 and Table S1. The monolayer VOI₂ structure comprises edge-sharing VO₂I₄ octahedra, where oxygen atoms are connected along the x -axis and iodine atoms are connected along the y -axis. As shown in Figure 3a, the FE phase emerges when V ions shift from their centrosymmetric positions within the octahedra along the x -direction, resulting in spontaneous $\mathbf{P}$. This symmetry breaking is driven by a pseudo-Jahn–Teller effect arising from hybridization between the unoccupied V 3d orbitals and occupied O 2p orbitals.³⁴ Notably, a Peierls transition occurs along the y -axis within the undistorted phase, inducing V–V dimerization, which further stabilizes the distorted phase³⁷ in Figure 3b. This distinctive dimerization, experimentally observed in NbOCl₂⁴³ and MoOCl₂,⁴⁴ provides a perfect material platform for our model.

We now examine the spin bands of monolayer VOI₂ using the exchange operation approach,⁴⁵ a well-established method for studying momentum-dependent spin splitting in AM.^{26,46} Ignoring magnetism, undistorted VOI₂ possesses the symmetries of $\{E, t_y, M_y, M_z, \text{ and } C_{2x}\}$. When AFM order is considered, its symmetries reduce to $\{E, M_z\}$. The missing symmetries $\{t_y, M_y, \text{ and } C_{2x}\}$ then act as exchange operations that link the magnetic sublattices, with t_y in particular enforcing spin-degenerate bands throughout the Brillouin zone (Figure 3d). By contrast, in the distorted phase, V–V dimerization breaks the t_y symmetry, leaving $\{M_y, C_{2x}\}$ as the dominant exchange operations, which leads to the emergence of AM, as shown in the momentum-dependent spin splitting bands (Figure 3e). The spin degeneracy along the Γ –X direction originates from the M_y or C_{2x} symmetry, while along the Γ –Y direction, it arises from a combination of these symmetries with inversion symmetry I . However, for momentum along the S– Γ –S' direction, no symmetry operation can map them back onto themselves, resulting in spin splitting exclusively along this direction. Remarkably, when the $\mathbf{P}$ is switched between

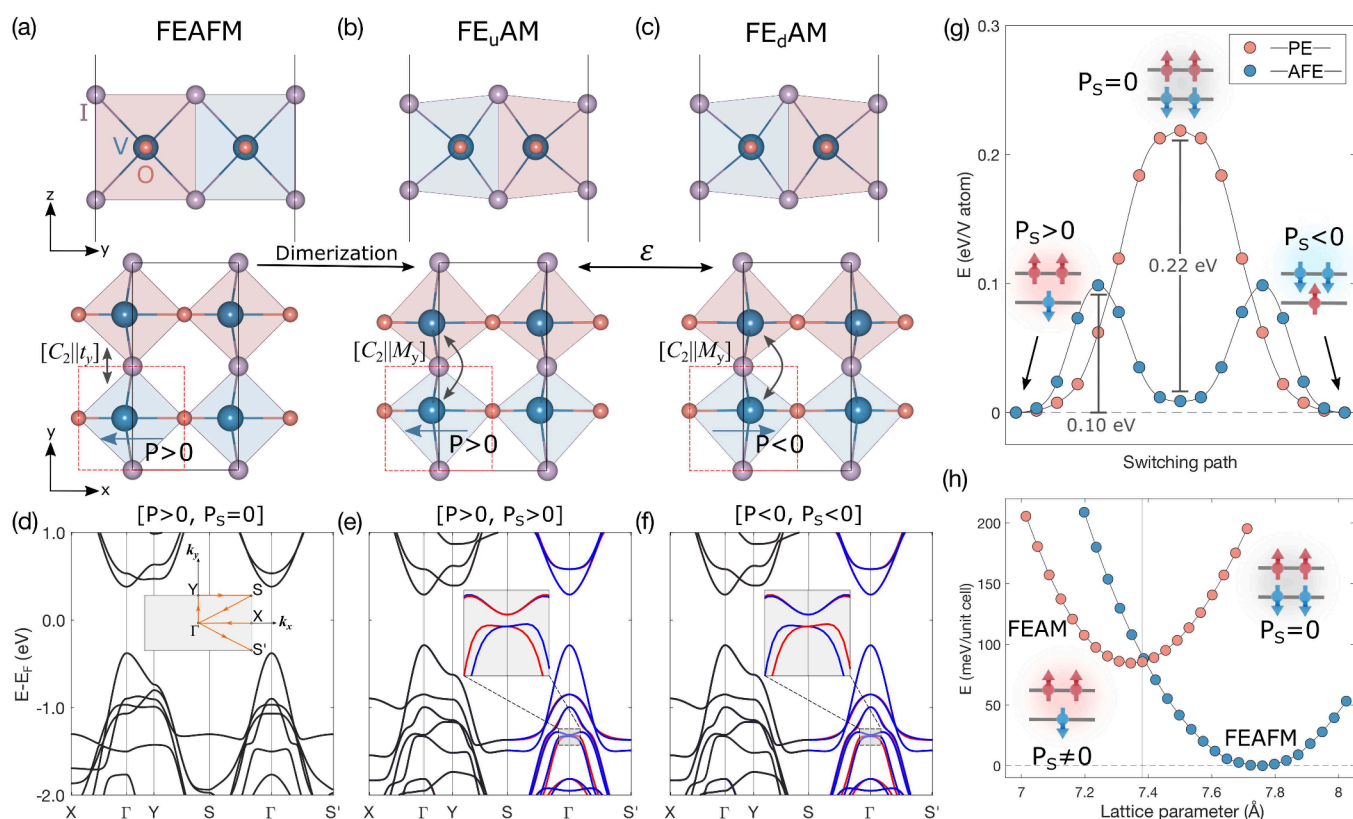


Figure 3. FEAM in 2D VOI₂ monolayers. (a) Structure of undistorted monolayer VOI₂ with $P > 0$. (b and c) Distorted configurations with $P > 0$ and $P < 0$, respectively. Red and blue polyhedra indicate opposite spin sublattices. (d–f) Calculated bands for panels a–c, respectively, where black, red, and blue denote the spin-degenerate, -up, and -down bands, respectively. The inset in panel d illustrates the Brillouin zone with its high-symmetry points. (g) Calculated kinetic pathways for the phase transition between FE_uAM and FE_dAM through paraelectric (PE) and antiferroelectric (AFE) intermediate states, demonstrating P_s reversal. (h) Lattice constant-dependent energies of both undistorted and distorted monolayer VOI₂, revealing strain-tunable phase transitions between the FEA FM and FEAM phases.

opposite FE states (Figure 3b,c), the sign of the P_s is simultaneously reversed (Figure 3e,f), demonstrating a robust magnetoelectric effect in 2D VOI₂.

To assess the feasibility and robustness of P -controlled P_s switching, we evaluated the transition energy barriers between the FE_uAM and FE_dAM states. We consider both the paraelectric (PE) phase, where V atoms remain undisplaced along the x -axis, and the AFE phase, where V atoms exhibit the opposite displacement, as intermediate states (Figure 3g). The calculated barriers via the PE and AFE states are 0.22 and 0.10 eV per V atom, respectively, comparable to those of other 2D multiferroic materials.⁴⁷ Our symmetry analysis and band structure calculations reveal that both the PE and AFE states exhibit conventional AFM behavior, i.e., $[P = 0, P_s = 0]$, as shown in Figure S4. Therefore, during the P switching, P_s undergoes not only sign reversal but also on and off toggling.

To understand the difference between the undistorted and distorted VOI₂ structure, we compare their relaxed lattice constants and find only a slight 0.4 Å difference along the y -axis (Figure 3h). Applying tensile strain increases the V–V distances, potentially suppressing Peierls distortion and driving the system toward the FEA FM phase. This structural evolution aligns with a clear transition from $P_s \neq 0$ to $P_s = 0$. These results demonstrate that strain engineering in VOI₂ can serve as an effective strategy for controlling lattice distortions, particularly for achieving t symmetry breaking.

The range of our proposed 2D FEAM candidates can be greatly extended by replacing certain elements by its same

group elements. This is confirmed by our calculations showing that FEAM properties are maintained in other vanadium oxyhalides (VOCl₂ and VOBr₂) and sulfide halides (VSX₂, where X = Cl, Br, or I), as discussed in Figures S5 and S6. Besides the Peierls-induced dimerization in monolayer VOI₂, alternative mechanisms for breaking t symmetry in FEA FM systems, such as Jahn–Teller distortions (alternating long and short specific bonds)²⁷ and rotations of magnetic lattice units,²⁸ can also induce FEAM. Fundamentally, these mechanisms all achieve spin splitting through inequivalent intraspine hoppings, while spin reversal originates from the interchanges of hopping amplitudes between spin sublattices in two opposite ferroelectric states.

Our theoretical framework and effective model provide a unified description of these diverse mechanisms, highlighting their general applicability. Beyond the single-phase materials proposed here, 2D FEAM can also be realized using alternative approaches such as stacking or sliding in van der Waals heterostructures.^{30,48,49} Our single-phase strategy relies on spontaneous symmetry breaking intrinsic to the material driven by ionic displacements, thereby enhancing experimental feasibility and potential device integration. In contrast, a layer stacking/sliding strategy achieves FEAM through engineered magnetic ordering coupled with interlayer charge inhomogeneity.^{30,48,49} Although these approaches differ in their microscopic origins of ferroelectricity, they share identical symmetry requirements, demonstrating complementary routes toward realizing our FEAM design principle.

Our proposed FEAM and its P-reversed P_s can be experimentally distinguished through angle-resolved photoemission spectroscopy measurements.^{13,15,50} Additionally, orientation-constrained magneto-transport measurements can also serve as an effective identification method.^{14,51,52} Beyond the two methods discussed above, we propose an optical approach to probe spin reversal. The calculated spin texture of VOI₂ exhibits a distinct d-wave character, and the opposite spin textures observed in the FE_uAM and FE_dAM states further confirm the electric-field-switchable spin splitting (Figure 4a).

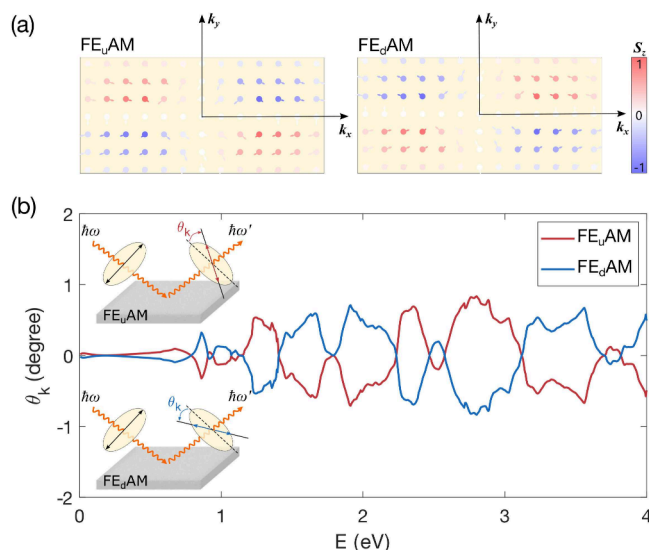


Figure 4. Probing the P_s reversal in FEAM. (a) Spin textures for FE_uAM and FE_dAM states in monolayer VOI₂, where in-plane spin components are indicated by arrows and out-of-plane components are represented by the color scale. (b) Calculated MOKE signals, characterized by Kerr angle θ_k . The inset schematically illustrates the MOKE measurement setup, with orange arrows showing the incident ($\hbar\omega$) and reflected ($\hbar\omega'$) light directions and black double-headed arrows depicting the polarization directions.

This reversal is further supported by MOKE measurements, which detect changes in the polarization of reflected light from magnetized surfaces, providing direct insights into magnetic properties.^{53–56} Figure 4b shows a prominent Kerr signal in the FE_uAM state due to anisotropic optical conductivity from time-reversal symmetry breaking. Crucially, the Kerr angle is correspondingly reversible with the polarization-dependent reversal of spin splitting in the FE_dAM phase.

In summary, our study establishes a universal, symmetry-guided framework for realizing 2D FEAM, addressing a significant challenge in integrating ferroelectricity with altermagnetism. By combining TB models with first-principles calculations, we demonstrate that carefully engineered lattice distortions can reconcile the symmetry requirements of both orders and induce the spin inequivalence essential for altermagnetic behavior. The identification of vanadium oxyhalides and sulfide halides as promising candidates not only validates our approach but also offers experimentally accessible platforms for testing these concepts. Furthermore, leveraging their 2D nature, our proposed FEAM can be integrated into van der Waals heterostructures, opening opportunities for proximity-induced phenomena and multifunctional device architectures.^{57–60} These advances underscore the urgent need for experimental realization and application-driven

studies to fully unlock the potential of 2D FEAM in next-generation, low-power spintronic, and multiferroic devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c02121>.

Details of the computational methods and additional information about the effective model, magnetic configurations, crystal and band structures for intermediate PE and AFE state VOI₂, and band structures for other 2D vanadium oxyhalides and sulfide halides (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Tong Zhou – Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 315200, China; orcid.org/0000-0003-4588-5263; Email: tzhou@eitech.edu.cn

Authors

Ziye Zhu – Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 315200, China; International Center for Quantum Design of Functional Materials (ICQD) and Hefei National Laboratory, University of Science and Technology of China, Hefei 230026, China

Xunkai Duan – Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 315200, China; School of Physics and Astronomy, Shanghai Jiao Tong University, Shanghai 200240, China

Jiayong Zhang – Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 315200, China; International Center for Quantum Design of Functional Materials (ICQD) and Hefei National Laboratory, University of Science and Technology of China, Hefei 230026, China; School of Physical Science and Technology, Suzhou University of Science and Technology, Suzhou 215009, China; orcid.org/0000-0001-7463-1751

Bowen Hao – Eastern Institute for Advanced Study, Eastern Institute of Technology, Ningbo, Zhejiang 315200, China

Igor Žutić – Department of Physics, University at Buffalo, State University of New York, Buffalo, New York 14260, United States

Complete contact information is available at:

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Author Contributions

[†]Z.Z., X.D. and J.Z. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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